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Ethics Surrounding The Health Inequality Project

Data collection often involves ethical considerations, especially when the aim is to analyze and address societal issues. The Health Inequality Project is a prime example of this.

According to the project's front page, its main goal is: "The Health Inequality Project uses big data to measure differences in life expectancy by income across areas and identify strategies to improve health outcomes for low-income Americans." This project aims to tackle the ethical issue of poor health among low-income Americans and, more broadly, to reduce the life expectancy gap between the rich and the poor.

While the project's goal is morally commendable and aligns with ethical principles, it is crucial for the founder to remain vigilant about the ethical implications of data collection, ownership, and processing. Despite its positive intentions, the project could inadvertently raise further ethical concerns.

Privacy and Confidentiality

The first issue to address is the hidden problem of privacy and confidentiality. Since the 1960s, the debate over privacy concerns related to data collection has become increasingly

prominent among Americans and globally (Igo 616; Bouk 627). This debate intensified after 2000 and accelerated around 2010 with the development of the internet, which facilitates the rapid spread of information and data, and the rise of artificial intelligence, which consumes vast amounts of data to build models. As highlighted by the New York Times article "The Battle for Digital Privacy Is Reshaping the Internet" (Chen), the rapid dissemination of information and the growth of artificial intelligence are transforming the world, making people more concerned about their data privacy than ever before.

Given this background of heightened privacy concerns and the public's desire to protect their data, the data collected by the project was scrutinized. According to the project's graph file and website, the program gathers various types of data, such as income amount, gender, household income percentile, and race (used for balancing purposes and later disaggregation) (Chetty). The data collected and processed by the project is at the individual level, so those managing the project must clearly define the boundaries of data collection before starting.

Information such as an individual's name, phone number, and Social Security Number (SSN) is not useful for the project and should not be collected. However, essential information like income should be collected. The collection of an individual's address should be approached with caution. While demographic information related to income levels and expected age depends on individuals' addresses and is vital for data analysis, overly detailed addresses could expose personal information. Therefore, the organization or individuals collecting this data must carefully consider the level of detail in the information they gather. They should ask, "How specific and detailed should this information be collected and presented to the public?" For address information, the collection was limited to the city level, ensuring that no more specific details about a person's address are disclosed beyond the city they reside.

However, after the data is collected, questions may arise about how the organization or personnel store and use the data. Although the data presented on the project's website is limited to a level that does not significantly impact the data provider's (the individual's) privacy, this is only what the project wants the viewer to see. It is difficult to determine whether the project's database contains more specific information. If it does, the privacy of the data providers becomes a concern again. These data could be leaked intentionally, such as through data trade, or unintentionally, such as through cyber-attacks and data breaches.

For instance, large firms like AT&T have experienced data breaches. Given the similarity between AT&T and this project in terms of widely collected individual data, a single data breach could result in a massive data leak. AT&T's recent data breach involved the exposure of 73 million customer records.

Even if the project does not store specific information about the data providers, the concern does not disappear. In the 21st century, with the advancement of the internet, the process of gathering and storing large quantities of data has become easier and more accessible. This means that databases are larger, leading to a greater affected area if the data is leaked. Among the 18 biggest data breaches of the 21st century, the smallest in terms of impact was the eBay data breach in 2014, which leaked 145 million records. The largest was the Yahoo data breach, which leaked up to 3 billion accounts in 2013 (with an additional 500 million leaked in 2014). All 18 massive breaches occurred after 2012, and some companies/organizations appeared twice (Yahoo and LinkIn) on the list. This implies that a severe data leakage that happened previously does not guarantee that the company/organization's data will not be breached again.

With the increasing frequency of massive data leaks, public concern about data security is high. Therefore, even if the Health Inequality Project does not collect specific data from the data providers, they still need to prove it. The project and its organizers should ensure transparency in how they utilize the data. This highlights the need for transparency in data storage and processing. However, transparency itself can lead to an ethical dilemma which limits the usage of this project.

Dilemma in Balancing Transparency and Bias

Before discussing the dilemma, it is important to examine the categories set by the project. One commendable aspect is that it separately counts the life expectancy of men and women. This approach aligns with a growing trend in statistics, as highlighted by the World Health Organization (WHO). In their article, WHO/Europe urges countries to collect gender data through their health information systems, stating, "Gender statistics help us to understand how the realities of life differ between the sexes, so that policy and services can be tailored to meet different needs" (WHO). This trend is also supported by articles from TIME and the World Bank Blogs. Counting women and children in statistics and handling this data with special care is now considered an ethical standard in data processing, which is reflected in the project's categories.

However, the project's mechanism for balancing data by race raises concerns. Although several studies have shown that life expectancy varies among races, and Dr. Barry A. Miller and his team have even found a connection between race and cancer risk (Miller), this approach carries the risk of supporting racism and creating stereotypes if the balancing standards are exposed. On the other hand, if the weighing mechanism lacks transparency, there is a higher risk

of the project organizers setting biased standards. This type of "backroom operation" occurs when transparency is lacking.

Similar to the race category, the New York Times article "Poor New Yorkers Tend to Live Longer Than Other Poor Americans" discusses how smoking and drinking are two factors that contribute to variations in life expectancy. However, delving into these data and using them to show connections with life expectancy in publicly available statistics could lead to the formation of stereotypes, such as labeling people in the Boston area as smokers or those in Chicago as heavy drinkers. The potential impact on the public should be carefully considered when forming these statistics.

The conflict here is that increased transparency could lead to public bias, while decreased transparency could result in organizational bias. It appears that the Health Inequality Project has reduced the number of categories and balancers used in their statistics, which aligns with the need for accountability.

Accountability

As previously mentioned, smoking and drinking, which are medically proven to affect life expectancy, were not included in the organization's statistical process, along with other factors such as drug usage, medical overdose, and medical insurance. Some might argue that excluding these factors makes the statistics vulnerable and weak in accountability since it does not consider as many variables as it could. However, this approach actually helps maintain basic accountability in their statistics.

So, how can accountability be maintained by reducing the number of factors involved in the statistics? The answer lies in the difficulty of quantifying these factors into "weights." These factors come from a wide range of areas, including government actions and personal habits, making it challenging to quantify them into specific weights that can balance the data and discuss the contribution of each factor. Forced quantification can result in the neglect of certain effects caused by specific factors, which decreases accountability even more compared to statistics that consider fewer factors. Like how Google's failure in their Flue trend statistics in 2013 (Reggio 252). Therefore, it is better to keep it simple.

These statistics are not intended to pinpoint specific factors that could shorten or extend life expectancy. Instead, they serve as indicators that show where life expectancy is higher or lower, providing a general path for further observations and analysis.

Conclusion

In summary, the project demonstrates a strong commitment to ethical considerations. It has taken significant steps to protect the personal information of data providers while making research results accessible by extracting the most useful information for statistical analysis, thereby safeguarding privacy and confidentiality. The project has also made efforts to maintain transparency while minimizing potential biases that could affect public perception. By limiting the factors involved in the statistics, the project has reinforced its accountability. These statistics are not intended to pinpoint specific factors that influence life expectancy but rather to serve as indicators of areas with higher or lower life expectancy, providing a general direction for further observations and analysis.

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